

CASE RECORDS of the MASSACHUSETTS GENERAL HOSPITAL

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Case 38-2022: A 21-Year-Old Woman with Fatigue and Weight Gain

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PRESENTATION OF CASE

Dr. Kathy M. Tran (Medicine): A 21-year-old woman was evaluated at this hospital because of fatigue, weight gain, and abdominal pain.

The patient had been well until 4 months before the current evaluation, when an episode of dull, aching pain in the right lower quadrant occurred. The pain lasted for a few hours before resolving spontaneously. During the next 4 months, episodes of abdominal pain continued to occur and increased in duration, frequency, and severity. The patient sought evaluation in the gynecology clinic of this hospital.

On evaluation, the patient described the pain as variable, rating it between 3 and 8 on a scale of 0 to 10, with 10 indicating the most severe pain. The episodes of pain occurred daily and lasted for up to 20 hours. The pain decreased with sitting or lying down and did not increase with movement; it was not associated with her menstrual period. The patient also reported fatigue, irritability, and weight gain of 4.5 kg in the previous 4 months despite increased exercise and decreased food intake.

There was a history of hypothyroidism, anxiety, irritable bowel syndrome, and migraines. The patient had never been pregnant and had no known history of sexually transmitted infections or abnormal results on screening tests for cervical cancer. She had previously been sexually active with male partners, but she reported no current sexual activity. She had taken oral contraceptives for 2 years but had stopped taking them 6 months before this evaluation, when a levonorgestrel-releasing intrauterine device (IUD) was placed. Other medications included levothyroxine, citalopram, and multivitamins; there were no known drug allergies.

The patient lived with roommates in an urban area of New England. She was a former athlete and worked as a research scientist. She did not smoke cigarettes, drink alcohol, or use illicit drugs. Her family history included hypothyroidism in her mother, coronary artery disease and hyperlipidemia in her father, breast cancer and diabetes in her paternal grandmother, and lung cancer in her paternal grandfather.

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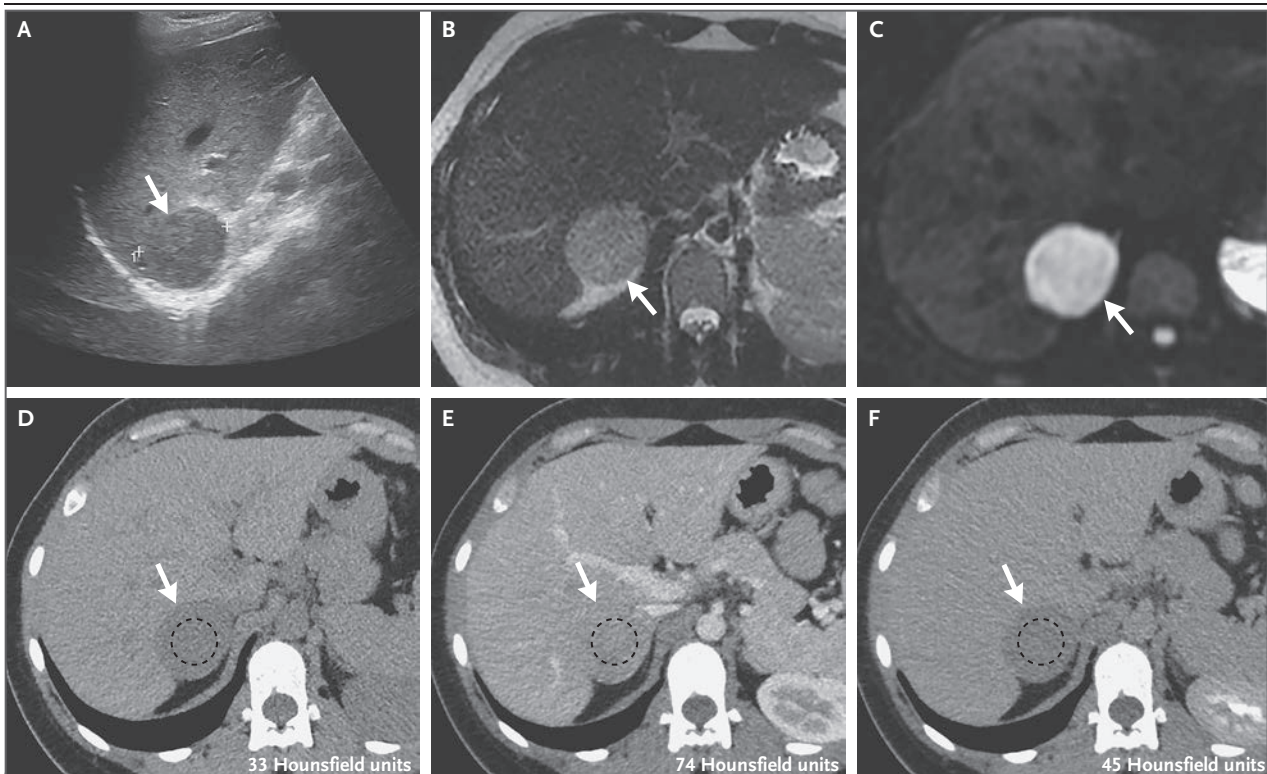


Figure 1. Imaging Studies.

A sagittal ultrasound image (Panel A) shows a hypoechoic mass (arrow) posterior to the right lobe of the liver. Magnetic resonance imaging shows a mass arising from the right adrenal gland that is isointense on a T2-weighted image (Panel B, arrow) and has diffusion restriction on a diffusion-weighted image (Panel C, arrow). Computed tomographic images obtained before the administration of contrast material (Panel D), during the portal-venous phase (Panel E), and during the 15-minute delayed phase (Panel F) show the density of the mass (arrows and dashed circles); the absolute washout is 71%.

On examination, the blood pressure was 119/73 mm Hg and the heart rate 62 beats per minute. The body-mass index (the weight in kilograms divided by the square of the height in meters) was 28.4. The thyroid was firm on palpation, with a small nodule noted on the right side. The abdomen was nondistended, soft, and nontender. A small hernia with a thrill was noted at the right inguinal ring. A speculum examination of the vagina and cervix was normal; IUD strings were visible at the cervical os. The complete blood count was normal, as were blood levels of electrolytes and glucose and results of tests for kidney, liver, and thyroid function. Screening tests for gonorrhea, chlamydia, human immunodeficiency virus, and syphilis were negative. Imaging studies were obtained.

Dr. Madeleine Sertic: Transabdominal and transvaginal ultrasonography of the abdomen and pel-

vis revealed a mass (measuring 4.8 cm by 5.0 cm by 3.9 cm) along the posterior aspect of the right hepatic lobe (Fig. 1A). The uterus, ovaries, and adnexa were normal. There was no free fluid in the pelvis, and the IUD was in an appropriate position.

Magnetic resonance imaging of the abdomen, performed before and after the administration of intravenous contrast material, revealed a right adrenal mass (measuring 4.5 cm by 4.7 cm by 4.0 cm) that was T2 isointense (Fig. 1B) and T1 hypointense. The mass had diffusion restriction (Fig. 1C), as well as early enhancement with washout, which refers to the temporal reduction in the enhancement of a lesion. The features were indeterminate and were not definitively diagnostic of adrenal adenoma or another adrenal lesion, such as pheochromocytoma or adrenocortical carcinoma.

Computed tomography (CT) of the abdomen and pelvis again revealed a right adrenal mass, which had a density of 33 Hounsfield units (HU) before the administration of contrast material (Fig. 1D), 74 HU during the portal-venous phase (Fig. 1E), and 45 HU during the 15-minute delayed phase (Fig. 1F). The absolute washout was calculated as 71%. The absolute washout is calculated by dividing the difference between the portal-venous-phase density and the delayed-phase density by the difference between the portal-venous-phase density and the unenhanced density and then converting the result to a percentage: $[(\text{HU portal-venous phase} - \text{HU delayed phase}) / (\text{HU portal-venous phase} - \text{HU unenhanced})] \times 100$. Adrenal adenomas tend to wash out faster than other adrenal lesions, and an absolute washout of more than 60% is highly suggestive of an adrenal adenoma. However, lesions larger than 4 cm are likely to be malignant, regardless of the washout characteristics.¹ There was also a thin-walled cyst (measuring 3.3 cm in diameter) in the lower lobe of the left lung.

Dr. Tran: The patient was referred to the endocrinology clinic of this hospital. Additional laboratory tests were performed to further evaluate the adrenal mass. Blood levels of renin and aldosterone were normal, as were urinary levels of catecholamines and metanephrines. An overnight, low-dose dexamethasone suppression test was performed. The baseline blood corticotropin level was 3 pg per milliliter (0.7 pmol per liter; reference range, 6 to 76 pg per milliliter [1.3 to 16.7 pmol per liter]). At 11 p.m., dexamethasone (1 mg) was administered; at 8 a.m., the blood cortisol level was 9.1 μg per deciliter (251 nmol per liter; reference value, <1.8 μg per deciliter [50 nmol per liter]), and the concurrent blood dexamethasone level was 631 ng per deciliter (reference range, 140 to 295). The 24-hour urinary free cortisol level was 31.0 μg (reference range, 3.5 to 45.0); the blood levels of dehydroepiandrosterone and dehydroepiandrosterone sulfate were normal.

Management decisions were made.

DIFFERENTIAL DIAGNOSIS

Dr. Allison Kimball: I participated in the care of this patient and am aware of the final diagnosis. When I initially evaluated this patient in the

endocrinology clinic, there were two key questions to address. First, was the adrenal mass benign or cancerous? Second, was the lesion hormone-secreting?

ADRENAL MASS

The differential diagnosis for the adrenal mass in this patient included adrenal adenoma (which is benign), pheochromocytoma, adrenocortical carcinoma, and adrenal metastasis. Imaging characteristics can be helpful in distinguishing among these possible causes, as well as in ruling out other entities. An adrenal cyst, adrenal hemorrhage, and myelolipoma (a benign tumor that consists of mature adipose tissue and hematopoietic components) could be ruled out on the basis of the imaging findings. Adrenal adenomas are typically round, homogeneous, and relatively small; the size of this patient's adrenal mass was atypical for an adrenal adenoma, but the homogeneous appearance was consistent with this diagnosis. Pheochromocytomas are typically highly vascular and may show cystic or hemorrhagic changes; neither of these features was seen in this patient. Adrenocortical carcinomas are typically unilateral and may be irregular in shape. Adrenal metastases are typically bilateral, which made this diagnosis unlikely in this patient.

The density of the mass before the administration of intravenous contrast material can be helpful in distinguishing among the various possible causes. Most adrenal adenomas have a high lipid content and thus have a low density on unenhanced CT (<10 HU). The density of this patient's adrenal mass on unenhanced CT was indeterminate (33 HU) and thus difficult to interpret.

The absolute washout after the administration of intravenous contrast material can also be helpful. A value of less than 50% at 15 minutes is characteristic of pheochromocytoma, adrenocortical carcinoma, and adrenal metastasis. A value of more than 60% at 15 minutes, which was seen in this patient, is most consistent with the diagnosis of adrenal adenoma.

ADRENAL ADENOMA

Given the imaging findings observed in this patient, the diagnosis of adrenal adenoma was suspected. The next step in her evaluation was to determine whether the suspected adenoma was

hormone-secreting, which would result in excess hormone levels. Primary aldosteronism (autonomous aldosterone secretion) leads to hypertension and, in some cases, hypokalemia; this patient's blood pressure and potassium level were normal. In addition, her aldosterone level was not elevated, and her renin level was not suppressed; these findings ruled out primary aldosteronism. Pheochromocytoma could be ruled out by the normal urinary levels of metanephrines and catecholamines.

Subclinical Cushing's syndrome (autonomous cortisol secretion without clinical manifestations of excess cortisol) is the most common hormonal abnormality diagnosed in patients with an incidentally noted adrenal adenoma. The patient's low baseline corticotropin level, in conjunction with a nonsuppressed cortisol level after the overnight administration of low-dose dexamethasone, was consistent with corticotropin-independent autonomous cortisol production. She did not have the physical findings associated with Cushing's syndrome, although she reported weight gain despite increased exercise, which could be related to autonomous cortisol production. The 24-hour urinary free cortisol level, which was normal in this patient, is not typically elevated in patients with subclinical Cushing's syndrome.

Overall, this patient's presentation was most consistent with a cortisol-secreting adrenal adenoma that was causing subclinical Cushing's syndrome. However, the size of the adrenal mass was worrisome. The risk of adrenocortical carcinoma was otherwise low, given her young age and the rarity of the condition. However, the risk of adrenocortical carcinoma is increased when the adrenal mass is greater than 4 cm in diameter.¹ Because of the need to rule out adrenocortical carcinoma, laparoscopic right adrenalectomy was performed.

DR. ALLISON KIMBALL'S DIAGNOSIS

Cortisol-secreting adrenal adenoma with subclinical Cushing's syndrome; adrenocortical carcinoma cannot be ruled out.

PATHOLOGICAL DISCUSSION

Dr. Baris Boyraz: The adrenalectomy specimen weighed 59 g and contained a mass (measuring

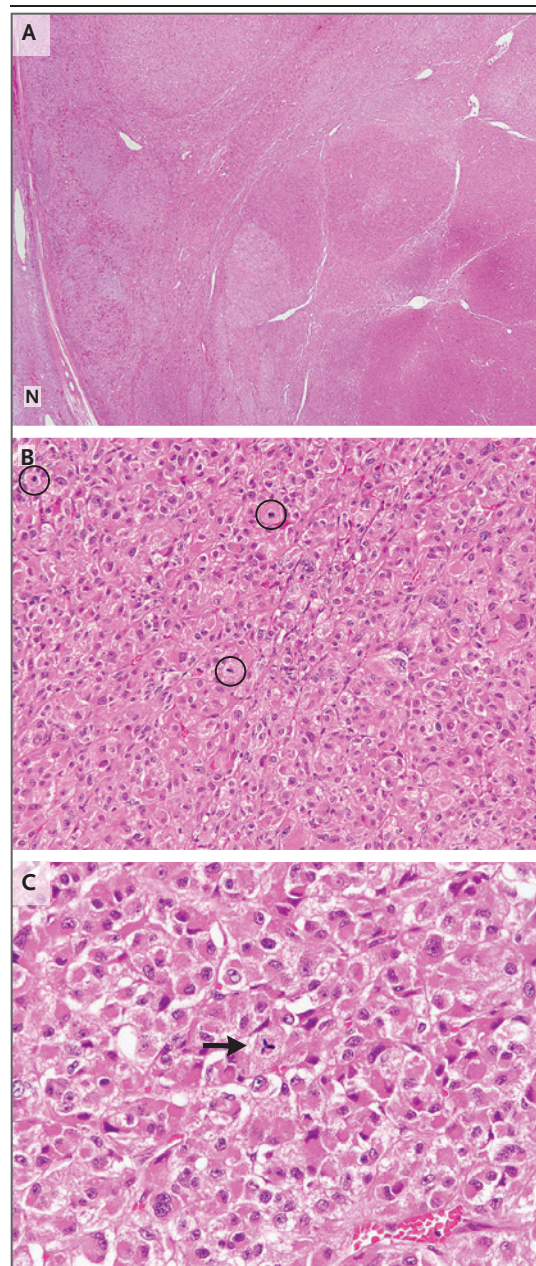


Figure 2. Specimen of the Adrenal Mass.

Hematoxylin and eosin staining shows a sharply circumscribed mass, abutting normal (N) adrenal gland, that has nodular architecture (Panel A). The mass is composed of oncocytic tumor cells that are growing diffusely in the nodules and have conspicuous mitotic activity (Panel B, circled). The tumor cells have rare, atypical mitoses (Panel C, arrow).

5.5 cm in greatest dimension) that was arising from the adrenal cortex. The mass had a golden-yellow cut surface and had a nodular growth

pattern, a finding that was also noted on microscopic examination (Fig. 2A). Tumor cells with abundant eosinophilic cytoplasm (oncocytic tumor cells) were growing diffusely both inside and between the nodules (Fig. 2B). They were remarkable for endocrine nuclear atypia, with enlarged, irregularly shaped nuclei and multinucleation. They also had increased mitotic activity, with up to 6 mitoses per 10 mm², including rare atypical mitotic figures (Fig. 2C). The Ki-67 proliferation index was 7%. Tumor was present focally at the inked resection margin.

The Lin–Weiss–Bisceglia criteria are used to determine whether oncocytic adrenocortical neoplasms are malignant. The presence of more than one major criterion is indicative of carcinoma. This patient had two major criteria: more than 5 mitoses per 10 mm² and atypical mitoses.^{2,3} Consequently, the neoplasm was diagnosed as adrenocortical carcinoma. The tumor was concluded to be of low grade because the level of mitotic activity was less than 20 mitoses per 10 mm². Overall, the pathological diagnosis was oncocytic adrenocortical carcinoma, low grade.

PATHOLOGICAL DIAGNOSIS

Oncocytic adrenocortical carcinoma, low grade.

DISCUSSION OF ONCOLOGY MANAGEMENT

Dr. Philip J. Saylor: Because the incidence of adrenocortical carcinoma is one case per 1 million persons,⁴ management is generally based on imperfect evidence. In the treatment of this patient, key considerations are her prognosis and the potential role of adjuvant systemic therapy or adjuvant radiation therapy (or both).

The patient had a pathological tumor stage of T2 and clinical node and metastasis stages of N0 and M0, respectively; thus, she had stage II disease. Cohort studies have suggested that, after tumor resection, stage II disease is associated with a chance of cure of approximately 70%.⁵ The patient's tumor was of a low grade, with 6 mitoses per 10 mm². The tumor Ki-67 proliferation index was 7%, which placed the patient in the best Ki-67 prognostic group.⁶

Outside clinical trials, adjuvant systemic ther-

apy for adrenocortical carcinoma primarily involves the use of mitotane, a derivative of the insecticide dichlorodiphenyltrichloroethane (DDT). The mechanism of action of mitotane is thought to be direct suppression of the adrenal cortex. Mitotane also alters the peripheral metabolism of steroids, which leads to numerous drug interactions.

The dosing of mitotane is highly individualized. Treatment usually begins with the administration of a low dose that is slowly increased while the serum drug level is measured and the patient is monitored for clinical side effects.⁷ Gastrointestinal, central nervous system, and endocrine side effects are particularly common. The elimination half-life of mitotane is long and variable.

The strongest evidence to support the use of adjuvant mitotane therapy comes from a retrospective analysis.⁸ Among patients with adrenocortical carcinoma who had undergone surgery, two cohorts received postoperative surveillance and one cohort received adjuvant mitotane therapy. Recurrence-free survival was longer in the mitotane cohort (42 months) than in the two observation cohorts (10 months and 25 months). These data are subject to the limitations of retrospective analyses.

One could make a case against the use of adjuvant mitotane therapy in this patient. She had a low Ki-67 proliferation index, and evidence from randomized, controlled trials to support the use of this therapy is lacking. In a recent randomized trial involving 91 patients who had adrenocortical carcinoma with a Ki-67 proliferation index of 10% or lower,⁹ mortality did not differ significantly between the observation group (5 deaths) and the mitotane group (2 deaths), but the results of this underpowered trial do not rule out the possibility of a benefit of mitotane.

This young and otherwise healthy woman had an expected 60 years of life ahead in the absence of a cancer recurrence. After we discussed the options with the patient and her parents, the patient favored treatment with adjuvant mitotane therapy. Our shared mindset was that the potential efficacy of the treatment outweighed the toxic effects, even if the evidence supporting the efficacy of mitotane for the treatment of this rare cancer was imperfect.

We planned for the patient to receive mito-

tane therapy for 2.5 years, a course similar to that used in the mitotane cohort of the retrospective analysis.⁸ We used a low threshold for discontinuation of therapy due to toxic effects. The patient was counseled to maintain effective contraception after the completion of treatment, until the serum mitotane level became undetectable.

Given the potential risk of locoregional recurrence after surgical resection, we referred the patient for a radiation oncology consultation before initiating mitotane therapy. If radiation therapy was to be pursued, it would be prioritized, and the mitotane dose would not be increased aggressively, given that the presence of bowel within the radiation field could lead to overlapping gastrointestinal side effects.

DISCUSSION OF RADIATION ONCOLOGY MANAGEMENT

Dr. Sophia C. Kamran: Local recurrence after surgical resection remains a common issue for patients with adrenocortical carcinoma, with estimated rates as high as 65%.¹⁰⁻¹³ Patients with features of advanced disease — such as a large tumor (>8 cm), positive resection margins, incomplete resection, a high Ki-67 proliferation index (>10%), rupture of the tumor capsule, or an advanced stage at diagnosis — are at a very high risk for locoregional recurrence.¹⁴

The role of adjuvant radiation therapy in the treatment of adrenocortical carcinoma has been debated, particularly because small, early studies suggested that adrenocortical carcinoma was relatively radioresistant, with adjuvant radiation therapy leading to poor local control rates and increased toxic effects.¹⁵ However, more recent case reports and retrospective analyses have suggested that adjuvant radiation therapy, with or without mitotane therapy, leads to a significant increase in local control.¹⁶ Several studies have suggested that adjuvant radiation therapy increases overall survival in patients with positive margins,¹⁷ whereas another study showed that, with doses of more than 40 Gy, adjuvant radiation therapy increases overall survival regardless of margin status and thus should be considered for select patients.¹⁵

For this young and otherwise healthy patient, there were multiple compelling reasons to rec-

ommend adjuvant radiation therapy. Her adrenalectomy had been performed with a laparoscopic approach, which yields a resection margin that is difficult to assess but is most likely positive. For the initial surgical management of adrenocortical carcinoma, laparoscopic adrenalectomy has been associated with higher rates of local recurrence and worse survival outcomes than open adrenalectomy. These outcomes were consistent even among patients who had a microscopically negative resection margin (R0).¹⁸ In addition, there was no evidence of distant disease on imaging after surgical resection in this patient.

The risks, potential benefits, and logistics of adjuvant radiation therapy were discussed with the patient and her family. The discussion emphasized the potential risks of pursuing adjuvant radiation therapy in the absence of data from randomized, controlled trials assessing the benefit of this treatment, owing to the rarity of the cancer. We reviewed the possible risks regarding renal function and fertility, as well as the risk of secondary cancer, which can have a profound effect on quality of life in a cancer survivor. Overall, the patient stated her preference for a proactive approach and elected to proceed with planning for radiation therapy.

Given the location of the tumor bed that was the target of irradiation, exposure of the right kidney to very high doses of radiation was unavoidable. Renal scintigraphy was performed to evaluate the function of the left kidney in case damage to the right kidney occurred; there was equal function in both kidneys. Because the patient was premenopausal, fertility counseling was provided before adjuvant radiation therapy was initiated, and the patient underwent oocyte cryopreservation.

We planned for the patient to receive an initial dose of 50.4 Gy (1.8 Gy per fraction) to a volume encompassing all areas with potential microscopic disease and lymphatic drainage, followed by a boost of 9.0 Gy (1.8 Gy per fraction) to a smaller volume encompassing the areas at highest risk for recurrence, for a total dose of 59.4 Gy in 33 fractions. Given this patient's young age, the goal was to minimize the dose to critical organs at risk and the excessive integral dose, because radiation therapy can confer a risk of secondary cancer in the future.¹⁹ We chose a

proton-based approach, because proton therapy can minimize radiation exposure to surrounding nontarget tissues, owing to the physical characteristics of the beam, including the Bragg peak, which results in energy deposition up to a finite depth in tissue.²⁰

In planning for radiation therapy, a four-dimensional CT simulation was performed to evaluate the level of motion in the target area that occurred with breathing (see video, available with the full text of this article at NEJM.org). The level of diaphragmatic breathing motion was 1.5 to 2.0 cm, which was borderline for proton-therapy acceptability, depending on the motion-mitigation strategies available at the institution. However, the target tumor bed was adjacent to the vertebral body; thus, with the use of posterior proton beams, a unique approach to administering a blended plan of gated photons and protons was thought to be achievable.

The risk of local recurrence in this patient was particularly high in the first 2 years after surgery, so we recommended that radiation therapy be initiated no later than 3 months after surgery.¹⁶ The patient started adjuvant radiation therapy 2 months after adrenalectomy. Although she received medication before each session of radiation therapy, she had nausea, which was treated with ondansetron. She completed radiation therapy successfully.

DISCUSSION OF ENDOCRINOLOGY MANAGEMENT

Dr. Kimball: Secondary adrenal insufficiency was anticipated to occur after adrenalectomy in this patient with Cushing's syndrome. High levels of cortisol exert increased negative feedback on the corticotrophs of the pituitary gland, thereby suppressing levels of corticotropin. This process can cause atrophy of the zona fasciculata of the unaffected adrenal gland, which leads to decreased cortisol production that becomes clinically apparent after adrenalectomy (Fig. 3A). Indeed, after adrenalectomy, the morning cortisol level was 0.5 μg per deciliter (reference range, 5 to 25), and the patient reported nausea and anorexia. Treatment with glucocorticoids was initiated, and nausea and anorexia resolved.

After beginning treatment with mitotane and radiation, the patient had nausea and fatigue. It

was difficult to determine whether the symptoms were associated with radiation, mitotane, or adrenal insufficiency, because these all can cause nausea and fatigue. The symptoms abated after the dose of glucocorticoids was increased to approximately two times the physiologic replacement level.

Mitotane inhibits steroidogenesis, and long-term use induces atrophy of the adrenal cortex (Fig. 3B). Atrophy of the zona fasciculata occurs after several weeks, leading to decreased cortisol production and adrenal insufficiency. Atrophy of the zona glomerulosa occurs in only some patients who are receiving mitotane and takes longer to develop, typically months. Atrophy of the zona glomerulosa results in aldosterone deficiency, which is treated with mineralocorticoid replacement. Because mitotane induces the 3A4 isoform of the cytochrome P-450 enzyme (CYP3A4), maintenance doses of glucocorticoids and mineralocorticoids that are several times as high as the usual doses are required.^{21,22} In most cases, adrenal insufficiency is reversible with discontinuation of mitotane therapy.²²

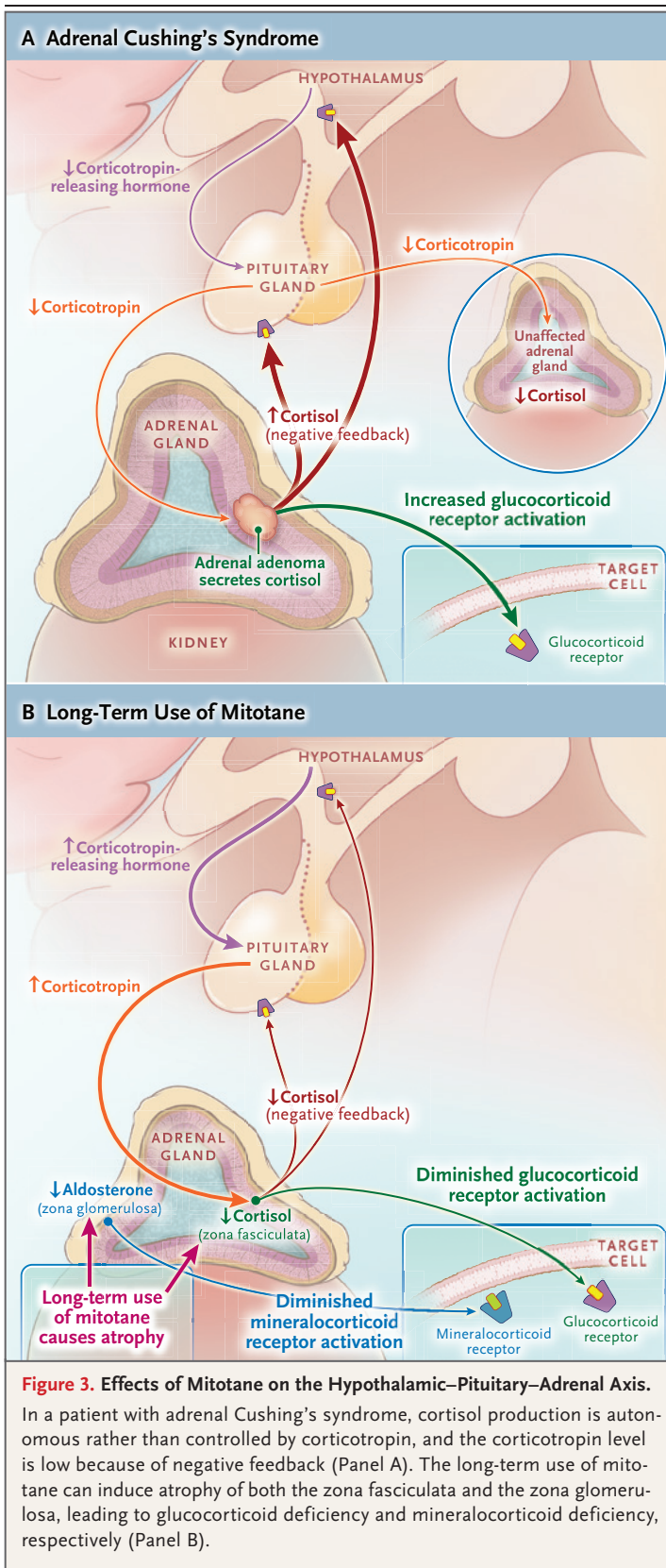
As the dose of mitotane was increased in this patient, fatigue, nausea, and anorexia occurred, which led to escalation of the dose of glucocorticoids. Three months after mitotane therapy was initiated, orthostasis developed. Orthostasis is suggestive of aldosterone deficiency, and it resolved after treatment with mineralocorticoids.

Mitotane can decrease levels of free thyroxine by increasing levels of thyroxine-binding globulin, and some evidence suggests that mitotane may directly inhibit the secretion of thyrotropin.^{4,23} The patient was taking levothyroxine for a preexisting diagnosis of hypothyroidism, and results of thyroid-function tests were monitored regularly and remained stable. Increased levels of follicle-stimulating hormone and luteinizing hormone have been observed in some premenopausal women who were receiving mitotane and may result in the development of ovarian cysts.²⁴

The patient continues to receive mitotane therapy, with serum mitotane levels ranging from 11.0 to 16.8 μg per milliliter. Imaging has shown no evidence of recurrent disease. One and a half years after diagnosis of the adrenocortical carcinoma, CT revealed a 9.5-cm adnexal cystic lesion with internal complexity. The patient un-



A video showing the four-dimensional CT simulation is available at NEJM.org



derwent right salpingo-oophorectomy, and pathological examination of the specimen revealed a mature cystic teratoma. Three months later, she was found to have a left adnexal cyst on routine follow-up imaging. On subsequent imaging, the left adnexal cyst had decreased in size from 5.0 cm to 1.6 cm.

PATIENT PERSPECTIVE

Patient: In the time since I received the diagnosis of adrenocortical carcinoma, I have become a cancer survivor, marathon runner, published neuroscientist, and prospective medical student. Nothing about this has been easy, but none of it would have been possible without the lifesaving multidisciplinary collaboration necessitated by the rarity of this disease.

There is never a good time to receive a cancer diagnosis, but the timing was especially shocking because I had just graduated from college, moved to a new city, and started my career in clinical research. When I presented to the clinic, and throughout most of my treatment, I was “young and healthy.” My lack of textbook clinical symptoms made it more difficult to diagnose my illness and distinguish me as a cancer patient.

I have become aware, however, that I am the expert on my own body, and at the time, this helped me to understand that something was not right for me. As an athlete, I was very in tune with my body and knew that it was crucial to advocate for myself in order to find the root cause of my symptoms. Notably, these symptoms could have easily been attributed to the change in routine I was also experiencing at the time. Luckily, my doctors genuinely listened to me and respected my persistence, which is one of the primary reasons I am still here today.

The extreme rarity of the condition, lack of literature, and unavailability of blood tests and biomarkers made it even more difficult to determine what was occurring biologically and how to approach treatment. This challenge has required collaboration across specialties, and I have been involved in each step of my disease experience. I value this lifesaving and unique approach, and I hope that a patient-first or individual-patient mentality is strongly incorporated into diagnosis, treatment, and overall patient care.

FINAL DIAGNOSIS

Adrenocortical carcinoma.

This case was presented at the combined Cancer Center and Pathology Grand Rounds.

Disclosure forms provided by the authors are available with the full text of this article at NEJM.org.

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